



ENME601 Thermodynamics

Week 5

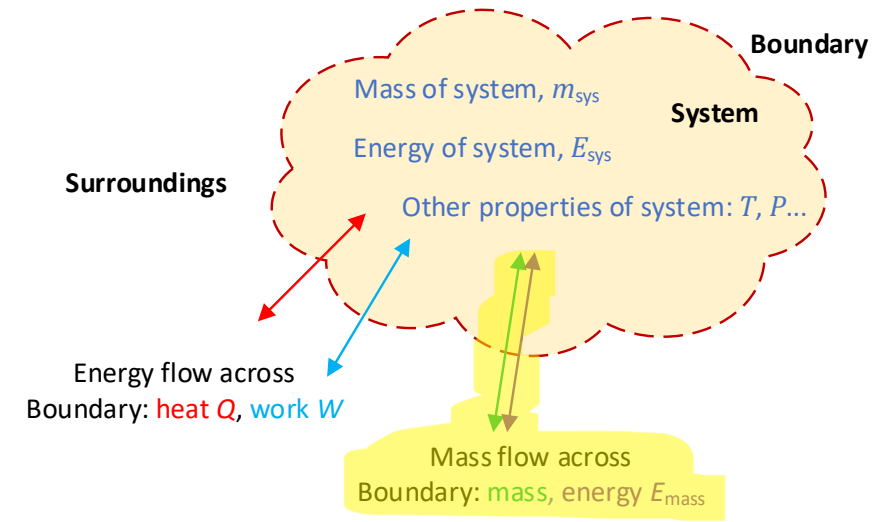
Presenter: Dr. Jay Wang

Objectives

- Develop the **conservation of mass principle** and apply it to steady flow control volumes.
- Apply the first law of thermodynamics as the statement of the **conservation of energy principle to steady flow control volumes**.
- Identify the **energy carried by a fluid stream** crossing a control surface as the sum of internal energy, flow work, kinetic energy, and potential energy of the fluid and to relate the combination of the internal energy and the flow work to the property enthalpy.
- Solve energy balance problems for **common steady-flow devices** such as nozzles, compressors, turbines, throttling valves, mixing chambers, heat exchangers and pipe & duct flow.

Introduction

- In the topic 5, we considered the first law energy balance applied to closed systems (no mass flow across boundary) and we considered the specific heat capacity and its relationship to internal energy and enthalpy for ideal gases, liquids and solids.
- In this topic we consider **open systems that include mass flow across the system boundary**
- We consider **steady flow systems** as a simplification of a more general open system.
- We develop and apply the **conservation of mass principle** and **conservation of energy principle** to **steady flow systems**.



Steady Flow Systems/Processes

- A large number of engineering devices (turbines, compressors, nozzles etc) operate for long periods of time under *steady conditions* i.e. **no changes with respect to time**. The *steady flow process* can model these devices.
- A steady flow process is a process where a *fluid flows through a control volume steadily*.
- Characteristics of steady flow system
 - **No properties** within the control volume **change with time**. (They may vary from point to point).
 - **Volume, mass and total energy content of the system (control volume) remain constant.**
 - **Properties** of mass flows in and out of the system remain **constant at the boundaries**.
 - **Heat and work interactions (across boundary) do not change with time.**



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Steady Flow Process/System

Steady = no change
with respect to time

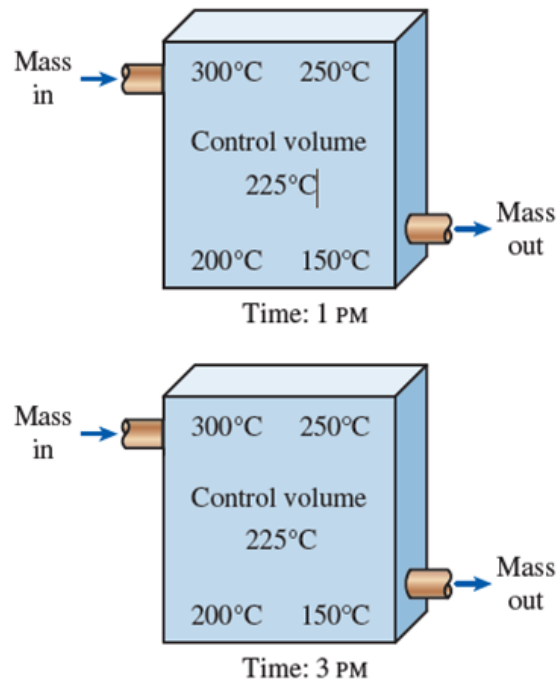


FIGURE 1-32

During a steady-flow process, fluid properties within the control volume may change with position but not with time.

Steady flow:
no change in mass or
energy inside system

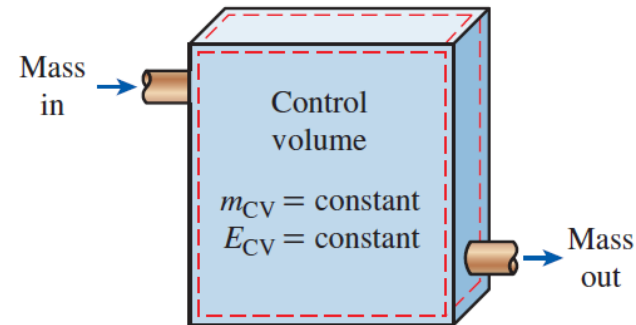


FIGURE 5-18

Under steady-flow conditions, the mass and energy contents of a control volume remain constant.

Steady flow:
Fluid properties at
inlets and exits do not
change with time

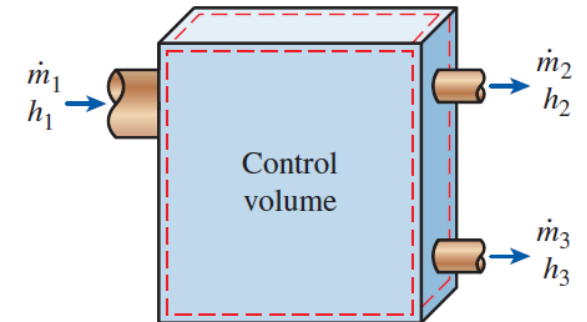


FIGURE 5-19

Under steady-flow conditions, the fluid properties at an inlet or exit remain constant (do not change with time).

Conservation of Mass Principle

Mass and Volume flow rates

Mass flow rate	Volume flow rate
$\dot{m} = \rho v_{\text{avg}} A \quad (\text{kg/s})$ $\dot{m} = \rho \dot{V} = \frac{\dot{V}}{v} \quad (\text{kg/s})$	$\dot{V} = v_{\text{avg}} A \quad (\text{m}^3/\text{s})$ Where: ρ = density of the fluid (kg/m^3) v_{avg} = average or bulk velocity in pipe/duct (m/s) A = cross-sectional area of pipe/duct (m^2) v = specific volume of fluid (m^3/kg)

Fluid flow:

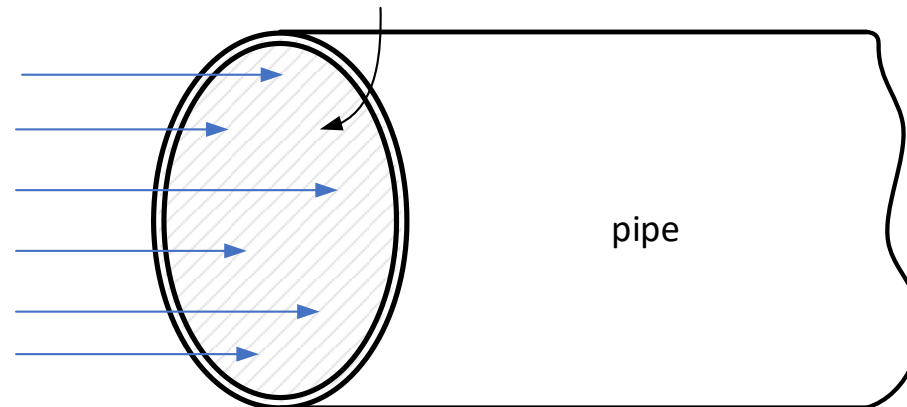
density, ρ

velocity, v

mass flow rate, $\dot{m}$

volume flowrate, $\dot{V}$

cross sectional area, A



Conservation of Mass Principle

Mass like energy is a conserved property

- Generally, for an open system (and for our model of a thermodynamic system) the conservation of mass principle can be expressed as:

$$\left(\begin{array}{c} \text{total mass entering} \\ \text{system in } \Delta t \end{array} \right) - \left(\begin{array}{c} \text{total mass leaving} \\ \text{system in } \Delta t \end{array} \right) = \left(\begin{array}{c} \text{net change in mass} \\ \text{in system in } \Delta t \end{array} \right)$$

Or
$$\sum \dot{m}_{\text{in}} - \sum \dot{m}_{\text{out}} = \Delta \dot{m}_{\text{system}}$$

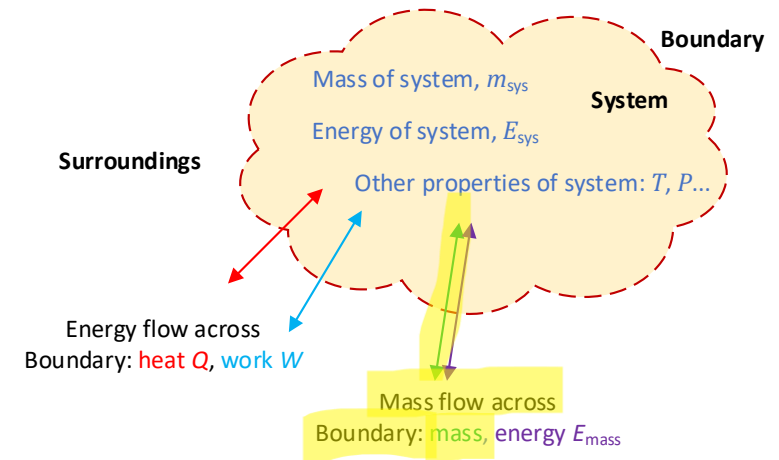
- In rate form this becomes:

$$\left(\begin{array}{c} \text{total mass flow} \\ \text{entering system} \end{array} \right) - \left(\begin{array}{c} \text{total mass flow} \\ \text{leaving system} \end{array} \right) = \left(\begin{array}{c} \text{rate of change of} \\ \text{mass in system} \end{array} \right)$$

Or
$$\sum \dot{m}_{\text{in}} - \sum \dot{m}_{\text{out}} = dm_{\text{system}}/dt$$

- For a **steady flow process the mass in the control volume is constant** so $dm_{\text{system}}/dt = 0$ and so

$$\sum \dot{m}_{\text{in}} = \sum \dot{m}_{\text{out}}$$



Mass Balance for Steady Flow Systems/Processes

- For a **steady flow system/process** the total amount of **mass contained in the control volume remains constant**, so total mass flow into control volume is equal to total mass flow out of control volume:

$$\sum \dot{m}_i = \sum \dot{m}_e$$

Where subscript i = inlets and e = exits or outlets

- In many cases there is just one inlet (1) and one outlet (2) so

$$\dot{m}_1 = \dot{m}_2 ,$$

$$\rho_1 v_1 A_1 = \rho_2 v_2 A_2 ,$$

$$\frac{v_1 A_1}{v_1} = \frac{v_2 A_2}{v_2}$$

Note: v = velocity (m/s)
 v = specific volume (m^3/kg) ($= 1/\rho$)

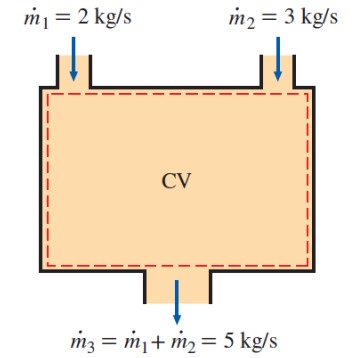


FIGURE 5–8

Conservation of mass principle for a two-inlet–one-outlet steady-flow system.

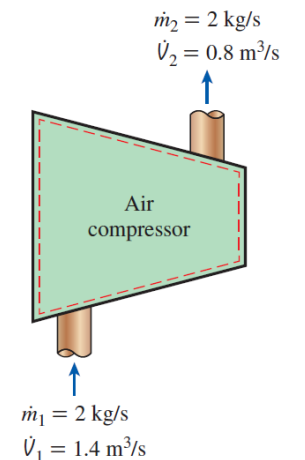


FIGURE 5–9

During a steady-flow process, volume flow rates are not necessarily conserved, although mass flow rates are.

Flow Work and the Energy of a Flowing Fluid

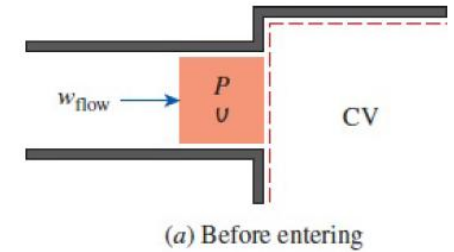
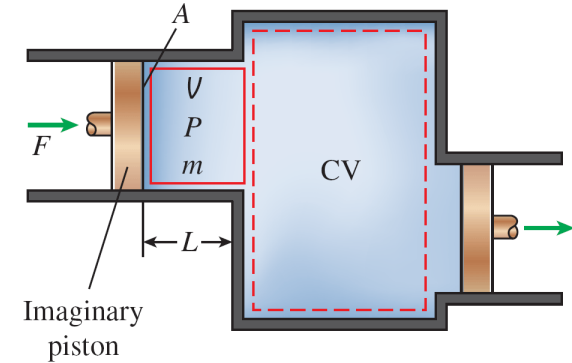
- **Flow work** is the work associated with pushing mass into or out of the control volume.
- Consider a fluid element of volume V and cross-sectional area, A . Fluid upstream at pressure P is pushing it into the control volume. (And force $F = PA$)
- The work done in pushing the volume element a distance L into the control volume is therefore:

$$W_{\text{flow}} = FL = PAL = PV \quad (\text{kJ})$$

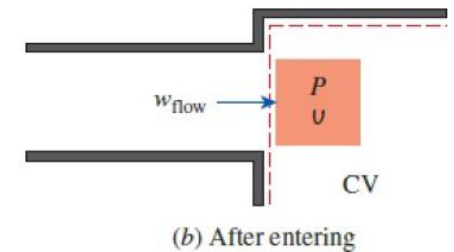
On a per unit mass: $w_{\text{flow}} = Pv \quad (\text{kJ/kg})$

Where v is the specific volume of the fluid

- Note: this is equivalent to the **flow energy**, that we discussed as a **component of the mechanical energy of a flowing fluid**.



(a) Before entering



(b) After entering

Flow Work and the Energy of a Flowing Fluid

- If we consider the small volume element in the previous slide as **not flowing** i.e. as a **normal system** with internal, kinetic and potential energies, then:

$$e_{\text{non-flow}} = u + \frac{v^2}{2} + gz \quad (\text{kJ/kg})$$

- To cause the volume element **to flow (in a pipe)** then we need to include the **flow energy Pv** to give the total energy of the flowing fluid:

$$e_{\text{flow}} = Pv + u + \frac{v^2}{2} + gz \quad (\text{kJ/kg})$$

And since $h = u + Pv$

$$e_{\text{flow}} = h + \frac{v^2}{2} + gz \quad (\text{kJ/kg})$$

- So, the **total energy flow rate, transported by mass** is

$$\dot{E}_{\text{mass}} = \dot{m}e_{\text{flow}} = \dot{m}(h + e_K + e_P) = \dot{m} \left(h + \frac{v^2}{2} + gz \right) \quad (\text{kW})$$

Flow Work and the Energy of a Flowing Fluid

- **Alternatively**, in week 1 we found that **the total specific mechanical energy** of a flowing fluid was made up of flow energy, kinetic energy and potential energy

$$e_{\text{mech}} = Pv + \frac{v^2}{2} + gz \quad (\text{kJ/kg})$$

- If we now include the **thermal energy component (u)** of the fluid, then the total energy of a flowing fluid is:

$$e_{\text{flow}} = u + Pv + \frac{v^2}{2} + gz \quad (\text{kJ/kg})$$

And since $h = u + Pv$

$$e_{\text{flow}} = h + \frac{v^2}{2} + gz \quad (\text{kJ/kg})$$

- So, the **total energy flow rate, transported by mass** is

$$\dot{E}_{\text{mass}} = \dot{m}e_{\text{flow}} = \dot{m}(h + e_K + e_P) = \dot{m} \left(h + \frac{v^2}{2} + gz \right) \quad (\text{kW})$$

Energy Analysis of Steady Flow Systems

- For a **steady flow process** the total energy of the control volume, remains constant i.e. the rate of change of energy of the control volume (system) = 0. Writing the energy balance equation in rate form:

$$\underbrace{\dot{E}_{\text{in}} - \dot{E}_{\text{out}}}_{\substack{\text{rate of net energy transfer} \\ \text{by heat, work and mass}}} = \underbrace{\cancel{\frac{dE_{\text{system}}}{dt}}^0}_{\substack{\text{rate of change in internal} \\ \text{kinetic, potential, etc energies}}}$$

- Thus, the rate of energy entering the control volume in all forms (heat, work and mass transfer) must be equal to the rate of energy leaving it

$$\underbrace{\dot{E}_{\text{in}}}_{\substack{\text{rate of net energy transfer} \\ \text{by heat, work and mass}}} = \underbrace{\dot{E}_{\text{out}}}_{\substack{\text{rate of net energy transfer} \\ \text{by heat, work and mass}}}$$

This can be written as:

$$\begin{aligned}\dot{Q}_{\text{in}} + \dot{W}_{\text{in}} + \sum \dot{E}_{\text{mass, in}} &= \dot{Q}_{\text{out}} + \dot{W}_{\text{out}} + \sum \dot{E}_{\text{mass, out}} \\ \Rightarrow \dot{Q}_{\text{in}} + \dot{W}_{\text{in}} + \sum \dot{m}_{\text{in}} e_{\text{flow, in}} &= \dot{Q}_{\text{out}} + \dot{W}_{\text{out}} + \sum \dot{m}_{\text{out}} e_{\text{flow, out}}\end{aligned}$$

Energy Analysis of Steady Flow Systems

$$\dot{Q}_{\text{in}} + \dot{W}_{\text{in}} + \sum \dot{m}_{\text{in}} e_{\text{flow,in}} = \dot{Q}_{\text{out}} + \dot{W}_{\text{out}} + \sum \dot{m}_{\text{out}} e_{\text{flow,out}}$$

- Note: e_{flow} is the total energy of a flowing fluid so $\dot{m}e_{\text{flow}}$ ($= \dot{E}_{\text{mass}}$) is the energy transported with mass flow to/from the control volume per unit time, So

$$\dot{Q}_{\text{in}} + \dot{W}_{\text{in}} + \underbrace{\sum \dot{m}_{\text{in}} \left(h_{\text{in}} + \frac{v_{\text{in}}^2}{2} + gz_{\text{in}} \right)}_{\text{for each inlet}} = \dot{Q}_{\text{out}} + \dot{W}_{\text{out}} + \underbrace{\sum \dot{m}_{\text{out}} \left(h_{\text{out}} + \frac{v_{\text{out}}^2}{2} + gz_{\text{out}} \right)}_{\text{for each outlet}}$$

- If we use the formal sign convention (heat transfer to a system and work done by a system are positive and heat transfer from a system and work done on a system are negative). And rearrange, then we can write this as:

$$\dot{Q} - \dot{W} = \underbrace{\sum \dot{m}_{\text{out}} \left(h_{\text{out}} + \frac{v_{\text{out}}^2}{2} + gz_{\text{out}} \right)}_{\text{for each outlet}} - \underbrace{\sum \dot{m}_{\text{in}} \left(h_{\text{in}} + \frac{v_{\text{in}}^2}{2} + gz_{\text{in}} \right)}_{\text{for each inlet}}$$

Energy Analysis of Steady Flow Systems

$$\dot{Q} - \dot{W} = \underbrace{\sum \dot{m}_{\text{out}} \left(h_{\text{out}} + \frac{v_{\text{out}}^2}{2} + gz_{\text{out}} \right)}_{\text{for each outlet}} - \underbrace{\sum \dot{m}_{\text{in}} \left(h_{\text{in}} + \frac{v_{\text{in}}^2}{2} + gz_{\text{in}} \right)}_{\text{for each inlet}}$$

General
form of
SFEE

- For a **single stream** (one inlet, one outlet), steady flow process then subscript 1 represents the inlet and subscript 2 represents the outlet and:

$$\dot{Q} - \dot{W} = \dot{m} \left[h_2 - h_1 + \frac{v_2^2 - v_1^2}{2} + g(z_2 - z_1) \right] \quad (\text{kW})$$

or more simply:

$$\dot{Q} - \dot{W} = \dot{m} [\Delta h + \Delta e_K + \Delta e_P] \quad (\text{kW})$$

SFEE: one inlet,
one outlet

In some cases we can ignore the changes in kinetic and potential energy so:

$$\dot{Q} - \dot{W} = \dot{m} \Delta h \quad (\text{kW})$$

- These steady flow energy equations are often referred to as SFEE's



Application of the steady flow energy equation (SFEE) to various engineering devices

Steady Flow Engineering Devices

Nozzles and Diffusers

- Nozzle = a device that **increases the velocity of a fluid** at the expense of pressure
- Diffuser = a device that **increases the pressure of a fluid** by slowing it down
- SFEE, single inlet & single outlet: $\dot{Q} - \dot{W} = \dot{m}[\Delta h + \Delta e_K + \Delta e_P]$

$\dot{Q} \cong 0$	velocities are high, so high speed process and heat does not have time to transfer, so essentially adiabatic
$\dot{W} = 0$	no forms of work cross the boundary (shaft, electrical etc)
$\Delta e_P \cong 0$	usually minor changes in elevation for a nozzle
$\Delta e_K > 0$	velocity changes are significant so we can't ignore kinetic energy changes

- SFEE for nozzles and diffusers

$$(h_2 - h_1) = - \left(\frac{v_2^2 - v_1^2}{2} \right) \quad \text{Nozzles and Diffusers}$$

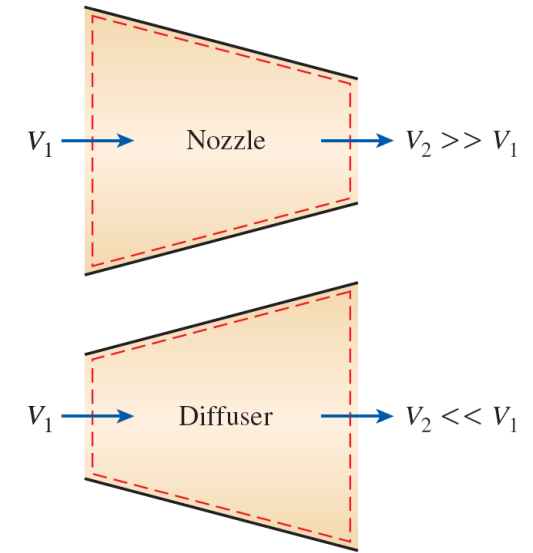


Figure 6–26: Nozzles and diffusers are shaped so that they cause large changes in fluid velocities and thus kinetic energies.



Jet engine diffuser

Steady Flow Engineering Devices

Turbines and Compressors

- Turbine = **produces work** (power) to drive generators, machinery, marine power. Can be steam or gas.
- Compressor = used to **increase the pressure of a gas**
- SFEE, single inlet & single outlet: $\dot{Q} - \dot{W} = \dot{m}[\Delta h + \Delta e_K + \Delta e_P]$

$$\dot{Q} \cong 0$$

Usually small relative to work produced or consumed. If intentional cooling in a compressor then $\dot{Q} \neq 0$.

$$\dot{W} > 0$$

Turbines produce work. Compressors consume work.

$$\Delta e_P \cong 0$$

usually minor changes in elevation for a turbine or compressor

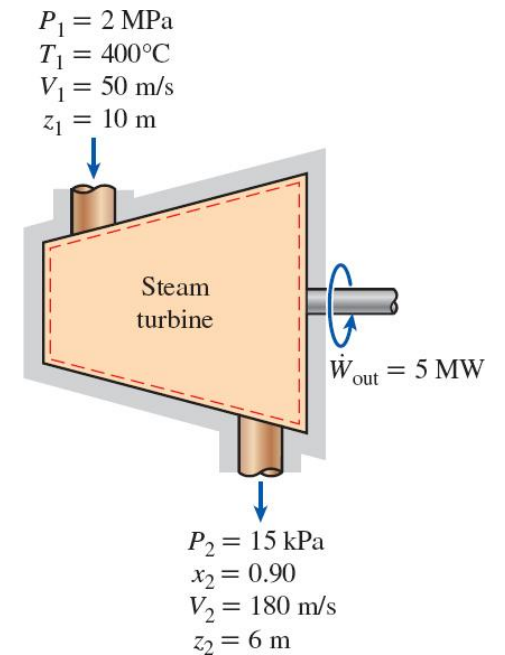
$$\Delta e_K \cong 0$$

velocities can be high in turbines (much lower in compressors) but Δe_K small relative to Δh .

- SFEE for turbines and compressors (simplest form)

$$\dot{W} = -\dot{m}\Delta h$$

Turbines and
Compressors



Turbine blades

Steady Flow Engineering Devices

Throttling valves

- These are flow restricting devices used principally to **reduce the pressure** in a flow stream. E.g. valve, porous plug, capillary tube
- SFEE, single inlet & single outlet: $\dot{Q} - \dot{W} = \dot{m}[\Delta h + \Delta e_K + \Delta e_P]$

$\dot{Q} \cong 0$	Small devices, with high velocities so essentially adiabatic
$\dot{W} = 0$	No work involved
$\Delta e_P \cong 0$	usually minor changes in elevation for a throttling valve
$\Delta e_K \cong 0$	Change in velocity can be high in throttles but Δe_K small relative to Δh .

- SFEE for throttles

$$0 = \dot{m}\Delta h \Rightarrow \Delta h = 0$$

so

$$h_1 = h_2 \quad \text{Throttle}$$

- Throttles are constant enthalpy or isenthalpic devices

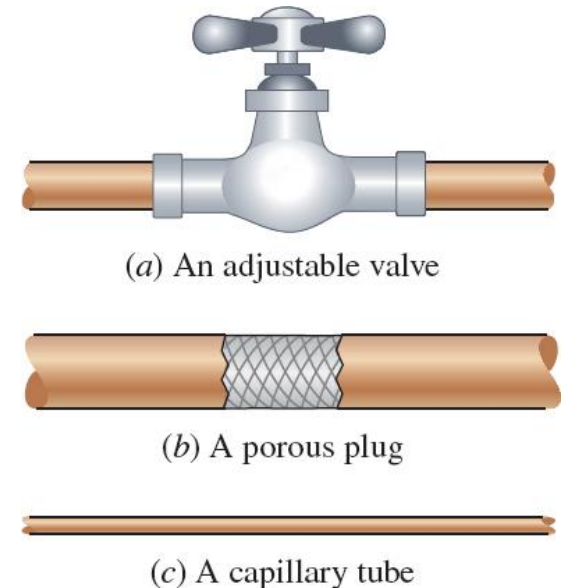


Figure 6–32: Throttling valves are devices that cause large pressure drops in the fluid.

Steady Flow Engineering Devices

Mixing Chambers

- The mixing of two or more streams of fluid at different temperatures, specifically **to achieve an intermediate temperature**.
- SFEE, multiple inlets & single outlet (general form):

$$\dot{Q} - \dot{W} = \underbrace{\sum \dot{m}_{\text{out}} \left(h_{\text{out}} + \frac{v_{\text{out}}^2}{2} + gz_{\text{out}} \right)}_{\text{for each outlet}} - \underbrace{\sum \dot{m}_{\text{in}} \left(h_{\text{in}} + \frac{v_{\text{in}}^2}{2} + gz_{\text{in}} \right)}_{\text{for each inlet}}$$

$\dot{Q} \cong 0$	Often insulated and high velocities so essentially adiabatic
$\dot{W} = 0$	No work involved
$\Delta e_p \cong 0$	usually minor changes in elevation for a mixing chamber
$\Delta e_K \cong 0$	Some changes in velocity but Δe_K small relative to Δh .

- SFEE for a mixing chamber

$$\underbrace{\sum \dot{m}_{\text{in}} h_{\text{in}}}_{\text{for each inlet}} = \underbrace{\sum \dot{m}_{\text{out}} h_{\text{out}}}_{\text{for each outlet}}$$

Mixing
Chamber

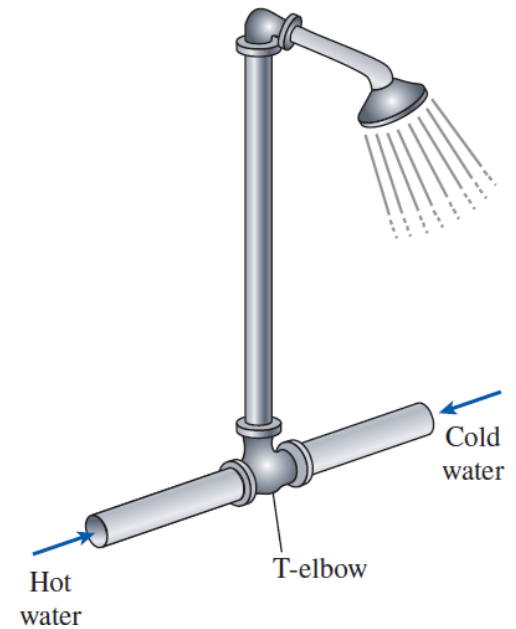


Figure 6–35: The T-elbow of an ordinary shower serves as the mixing chamber for the hot- and the cold-water streams.

And also apply a mass balance
$$\sum \dot{m}_{\text{in}} = \sum \dot{m}_{\text{out}}$$

Steady Flow Engineering Devices

Heat Exchangers

- A **hotter fluid exchanges heat with a cooler fluid**. The fluids are kept separate by a wall or pipe etc.
- SFEE, single inlet & single outlet (consider hot and cold streams separately): $\dot{Q} - \dot{W} = \dot{m}[\Delta h + \Delta e_K + \Delta e_P]$

$\dot{Q} > 0$	Heat lost by hotter fluid = heat gained by cooler fluid
$\dot{W} = 0$	No work involved
$\Delta e_P \cong 0$	Usually minor changes in elevation for a heat exchanger
$\Delta e_K \cong 0$	Some changes in velocity but Δe_K small relative to Δh .

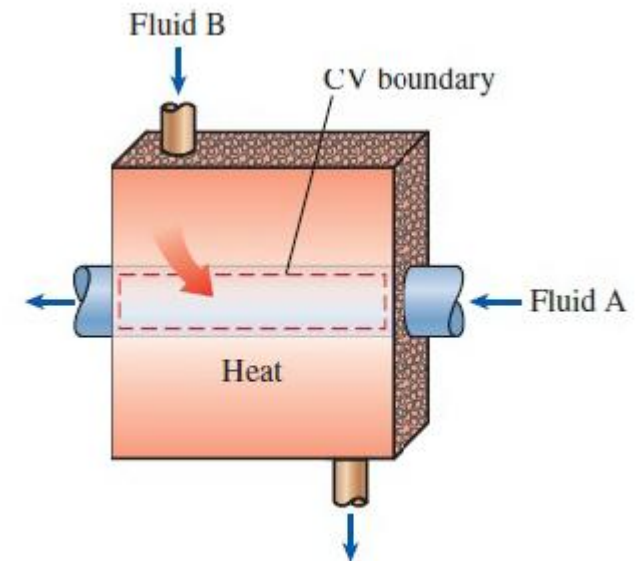
- SFEE's for a heat exchanger

$$\dot{Q}_{\text{cold}} = (\dot{m}\Delta h)_{\text{cold}} = [\dot{m}(h_{\text{out}} - h_{\text{in}})]_{\text{cold}}$$

$$\dot{Q}_{\text{hot}} = (\dot{m}\Delta h)_{\text{hot}} = [\dot{m}(h_{\text{out}} - h_{\text{in}})]_{\text{hot}}$$

$$-\dot{Q}_{\text{hot}} = \dot{Q}_{\text{cold}}$$

Heat
Exchanger



(b) System: Fluid A ($Q_{CV} \neq 0$)

Heat loss by hot fluid =
heat gain by cold fluid

Steady Flow Engineering Devices

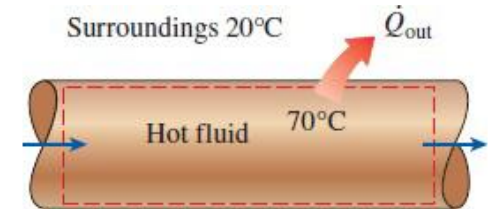
Pipe and Duct flow

- All forms of energy may be involved in the flow of fluids in pipes and ducts, heat transfer, work interactions and changes in kinetic and potential energies
- SFEE, single inlet & single outlet: $\dot{Q} - \dot{W} = \dot{m}[\Delta h + \Delta e_K + \Delta e_P]$

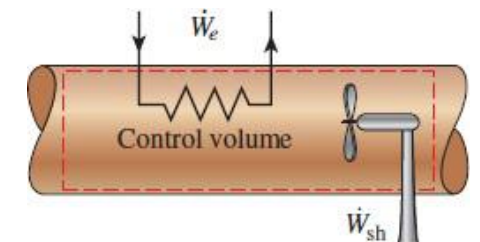
$\dot{Q} > 0$	heat transfer to or from the fluid: boiler tubes, refrigeration coils, duct heaters etc.
$\dot{W} > 0$	if control volume contains a heater ($\dot{W}_e$) or a fan or pump ($\dot{W}_s$) then work interactions do take place.
$\Delta e_P > 0$	pipes and ducts usually undergo elevation changes so potential energy changes could be significant.
$\Delta e_K \cong 0$	if $\dot{Q}$ is significant and there are no large diameter changes then changes in kinetic energy can be insignificant.
- SFEE for pipe and duct flow (no simplification)

$$\dot{Q} - \dot{W} = \dot{m}[\Delta h + \Delta e_K + \Delta e_P]$$

Pipe & Duct
Flow



Heat transfer



Work interactions